

Fundamental Information for Low-Turnover Equity ML: Evaluating a Mamba-Inspired State Space Model for Long-Horizon Stock Trading*

Terry Ding

October 2026

Abstract

Stock price prediction with modern neural networks is a growing field that attracts the attention of both investors and scholars, yet most research is oriented toward short-term prediction, where models target price movements only a few days ahead. This paper departs from that direction: it uses fundamental stock data to train a Mamba-Inspired State Space (MISS) model that generates a low-turnover, sector-neutral investment strategy. The study uses Yahoo Finance OHLCV data, SEC filings, and custom calculated technical indicators, and compares MISS against LSTM, StockMixer, and graph neural network architectures. Across five out-of-sample years (2021–2025), the long-horizon fundamental strategy records a mean annual return of 17.53% and a mean annualized Sharpe ratio of 1.104, with approximately 32 portfolio trade events per year. The matched long-horizon technical strategy records -6.84% and -0.299 , while the short-horizon technical strategy records -4.48% and -0.276 with approximately 605 trade events per year. A 10,000-resample moving-block bootstrap assigns probability 0.988 to a positive Sharpe difference between the fundamental and short-horizon technical strategies, and the fundamental strategy is the only one of the twelve evaluated configurations whose Deflated Sharpe Ratio remains at approximately 1.0 after multiple-testing adjustment. The results indicate that fundamental information

becomes substantially more useful when the prediction horizon matches the slower rate at which company fundamentals change, although the effect remains dependent on market regime and model architecture.

Keywords: financial machine learning; fundamental data; state space models; Mamba; long-horizon trading; walk-forward validation; transaction costs; sector neutrality.

1 Introduction

The goal of accurately predicting stock prices has been the core purpose of financial machine learning research for decades. This research is valuable to investors, as better predictions can help create more robust and consistent investment strategies. Neural networks are particularly useful for achieving this task, as their ability to capture non-linear trends is key for navigating the complex space of stock markets. That is why individual researchers along with large asset managers and funds are increasingly using Artificial Intelligence (AI) to generate accurate stock price signals [5, 7]. To accurately evaluate stock prediction methodologies, it is necessary to analyze each underlying variable. Indeed, AI has already transformed the financial sector by automating tasks like fraud detection, risk assessment, credit scoring, and algorithmic trading. A growing sector of this field is stock price prediction (SPP), where models are directly used to predict future stock returns or price movements. AI is employed for this task because of its ability to recognize patterns in data that humans normally would not be able to. For example, Fischer and Krauss demonstrated that deep LSTM networks can generate statistically significant daily return signals

*Reproduction edition, October 2026. Every empirical result reported here is from an independent end-to-end reproduction of this study; the training pipeline, backtest code, metrics, and figure sources are available at https://github.com/ShadowKingYT444/Fundamental_SPP. The relationship between these results and the values reported previously is discussed in Section 5.

across a universe of S&P 500 constituents over a long out-of-sample period [5].

Stock prices are not just related to the company; a variety of factors influence prices. Financial machine learning models typically ingest fundamental, technical, and sentimental factors. Fundamentally, the company’s earnings, products, balance sheet, and broader economic indicators like interest rates and inflation all affect how inherently valuable a company is. On a technical level, market dynamics like supply and demand, liquidity, volume, volatility, and a variety of trading patterns can move stock prices in the short term. Finally, at a sentiment level, news events like wars, political decisions, and government policies may affect investor psychology, driving price movement. To evaluate investment strategies, metrics like the Sharpe Ratio reveal how stable an investor’s strategy is by comparing excess return against the volatility needed to achieve that return.

Leveraging AI to accurately predict the future value of a stock presents a number of challenges that must be overcome. Stock datasets inherently contain a large amount of noise that models must learn to filter out while retaining useful variables [8]. The complex nature of how stock markets behave means that while a variety of factors affect a stock price, those factors can change at any moment. The amount of time-series data for an individual stock is also inherently limited, as there are only around 250 trading days a year. However, recent research has still shown that nonlinear ML models can identify economically useful signals from large stock panels [5, 7].

Currently, the majority of stock-prediction research has been oriented towards short-term price prediction, and many papers still evaluate models primarily with statistical prediction metrics instead of deploying their signals into simulated portfolios. This makes it difficult to tell whether a small improvement in prediction actually becomes a better investment strategy after turnover and transaction costs are included [5, 2]. Thus, we present a novel investment framework that leverages SEC fundamental data within a Mamba selective state space model to generate low-turnover, sector-neutral signals evaluated across five out-of-sample years using walk-forward validation and robustness diagnostics. To understand whether the gains come from the data or only from the architecture, the same exper-

iment is also conducted with long-horizon technical data and short-horizon technical data, and MISS is compared against LSTM, StockMixer, and GNN models. To evaluate results, we leverage portfolio metrics like annual return, Sharpe ratio, turnover, maximum drawdown, transaction-cost sensitivity, block bootstrap testing, and the Deflated Sharpe Ratio framework [1].

The rest of this paper is organized as follows: Section 2 offers a brief literature review of the financial machine learning (ML) field. Section 3 presents the method proposed. Section 4 showcases the results of the experiments. The final section concludes this paper and discusses future directions of research.

2 Literature Review

Significant progress has been made in the field of financial machine learning (ML), transitioning from fundamental feed-forward neural networks to complex architectures like the Transformer and State Space Models. This review highlights the architectural progress of the field while critiquing the current short-term technical bias and the lack of practical portfolio application.

2.1 Deep Learning in Financial Time Series

Basic approaches using simple feed-forward networks have been replaced with complex architectures designed to properly ingest complex time-series data. LSTM networks have been a main choice of model due to their capacity to retain historical state information through gates that selectively store or forget information over time [9]. By mitigating the vanishing gradient problem present in standard RNNs, LSTMs can capture longer-range temporal dependencies in market data [12]. Recently, the Transformer architecture has also been adapted for financial time series [15]. This model uses the self-attention mechanism to simultaneously weigh the relevance of previous steps, unlike LSTMs which process data sequentially. This approach allows the model to detect relationships between distant observations and between different features. However, standard Transformer attention introduces quadratic computational cost with sequence length, making it more expensive when a large amount of historical context is required.

More recent financial models have started exploring architectures beyond LSTMs and Transformers. StockMixer directly mixes information across indicators, time, and stocks through lightweight MLP blocks, allowing dense technical information to interact without a costly attention layer [3]. Graph Neural Networks (GNNs) take a different approach by treating stocks as connected entities rather than independent time series. Relational stock-prediction models have shown that industry links and other relationships between firms can improve cross-sectional ranking because information about one company may also contain information about connected companies [4, 11]. These models provide useful comparisons for MISS because they solve different parts of the same problem: temporal memory, cross-feature mixing, and cross-stock relationships.

2.2 Short-Term Technical Bias

The majority of recent literature relies heavily on daily or intraday OHLCV (Open-High-Low-Close-Volume) data combined with technical indicators; studies often target price movements merely a few days into the future [5]. This short-term technical bias means that even when models achieve useful directional accuracy, they can generate high-turnover signals that are vulnerable to market microstructure noise and transaction costs. Research on portfolio construction has demonstrated that transaction costs can materially change which stock characteristics remain useful after trading is accounted for [2]. Due to these costs, the need for a lower-turnover strategy becomes apparent. Research should therefore also consider market fundamentals that move prices over longer periods rather than only short-lived technical price moves [14].

2.3 Fundamental Data

Leveraging fundamental data such as SEC filings, earnings, cash flow, valuation ratios, and balance-sheet information within neural networks is a significantly sparser area of research. Fundamental data is harder for models to digest because it is inherently sparse, with many accounting variables released quarterly rather than daily, and it is subject to strict point-in-time constraints to avoid look-ahead bias. However, integrating these eco-

nomics indicators allows models to capture information that is more directly connected to the company itself. Gu, Kelly, and Xiu show that machine-learning methods can extract strong nonlinear relationships from large panels of firm characteristics, supporting the broader idea that accounting and market characteristics contain meaningful cross-sectional information [7]. In practice, any such pipeline must also ensure that a filing does not enter the data before it was actually made public; SEC EDGAR filing metadata makes this point-in-time alignment possible [13].

2.4 Mamba-Inspired State Space Models

The primary innovation of this paper is the application of Selective State Space Models (SSMs), specifically the Mamba architecture [6]. This newer framework attempts to resolve issues present in both Transformers and recurrent models. Mamba’s computational cost grows linearly with sequence length, as it processes a sequence through recurrent state updates instead of computing full pairwise self-attention. More importantly for this research, the architecture selectively changes how inputs affect the hidden state based on the current input. This creates a mechanism that can keep useful information while allowing irrelevant information to disappear. In financial data, this is especially useful because an earnings or balance-sheet update may stay important for months while individual day-to-day price fluctuations may not.

This does not mean that Mamba should automatically outperform every model. StockMixer is specifically designed to mix dense market features across time and stocks, while GNN models can directly represent relationships among firms. For this reason, the experiments compare architectures under the same portfolio construction rather than only comparing MISS to an older LSTM baseline. The purpose is to test whether selective state-space memory is particularly useful for sparse, long-horizon fundamental information instead of claiming that one architecture is universally superior.

2.5 Portfolio Construction vs. Point Prediction

Finally, a critical disconnect exists between model accuracy measured by metrics such as Mean

Squared Error (MSE) and actual trading profitability. A model can achieve a lower prediction error and still lose money if it is overfit, trades too frequently, or makes its largest mistakes during highly volatile periods. Financial ML is especially vulnerable to this problem because testing many models, factors, and hyperparameters can make a backtest appear significant by chance [8]. To address these issues, research has shifted towards using portfolio metrics and more advanced statistical controls. The Deflated Sharpe Ratio adjusts an observed Sharpe ratio for selection bias, non-normal returns, and the number of strategies or model configurations that were tested [1]. This paper therefore evaluates the complete portfolio rather than judging the model only from prediction error.

3 Methodology

3.1 Data and Feature Construction

The experiment uses three primary categories of information: market data, fundamental data, and custom technical indicators. Daily Open-High-Low-Close-Volume data is taken from Yahoo Finance [16], while fundamental accounting information is aligned using SEC filing data [13]. The fundamental feature set includes profitability measures such as return on assets and operating margin, growth variables such as revenue and earnings growth, valuation variables such as earnings and free-cash-flow yield, balance-sheet variables such as leverage and liquidity, and quality variables such as accruals. Technical features include multi-period returns, realized volatility, moving-average distance, RSI, MACD, ATR, volume surprise, and price-volume trend measures.

A major concern with fundamental data is look-ahead bias. A company’s quarter may end before the market actually has access to the corresponding filing. Therefore, fundamental values are only added to the dataset after the public filing acceptance date instead of being backfilled to the fiscal period end. Features are winsorized using thresholds calculated from the training data and standardized cross-sectionally on each date. This prevents very large outliers from dominating the model while also avoiding the use of statistics from the future.

The universe is the S&P 500. Because point-in-

time fundamental coverage is incomplete, the fundamental regime scores between 161 and 201 names per year, while the technical regimes score between 441 and 474 names per year. This coverage difference is a property of the data, not of the models, and it matters for interpreting portfolio-level results (Section 5).

3.2 Prediction Horizon and Target

The main hypothesis of this paper is that the usefulness of a feature depends on the horizon at which it is evaluated. Fundamental variables change slowly, so forcing them to predict the next one or two days may create an unnecessary mismatch between the data and the target. The main long-horizon task therefore predicts a 63-trading-day forward return, which is approximately one quarter. A shorter five-day target is used for the technical benchmark.

For stock i on day t , the main target is the stock’s forward return minus the average forward return of other stocks in its sector:

$$y_{i,t}^{(63)} = r_{i,t \rightarrow t+63} - \bar{r}_{s(i),t \rightarrow t+63}, \quad (1)$$

where $s(i)$ represents the stock’s sector. This makes the task closer to selecting better companies within an industry rather than simply predicting whether the entire technology or energy sector will rise.

3.3 Mamba-Inspired Selective State Model

For each stock, the daily feature vector is projected into a hidden representation and passed through stacked selective state-space blocks. At a simplified level, the model can be represented as

$$h_t = \bar{A}_t h_{t-1} + \bar{B}_t x_t, \quad (2)$$

$$z_t = C_t h_t + D x_t, \quad (3)$$

where x_t is the current input and h_t is the model’s running hidden state. Unlike a standard state-space model with the same transition at every step, Mamba allows important parameters to depend on the current input [6]. The practical intuition is that the model can learn that some observations should strongly update memory while others should mostly be ignored. This is the main reason the architecture is tested on sparse fundamental data.

The MISS implementation uses two selective state-space blocks, residual projections, normalization, and a final scoring head. The model is kept at

approximately 0.2 million trainable parameters so that performance gains cannot be explained only by giving it substantially more capacity than the baselines. Training combines a Huber regression loss with a pairwise ranking loss:

$$\mathcal{L} = \mathcal{L}_{\text{Huber}}(\hat{y}, y) + \lambda \mathcal{L}_{\text{rank}}(\hat{y}, y), \quad (4)$$

with $\lambda = 0.25$. The regression component learns the magnitude of future returns while the ranking component more directly teaches the model which stocks should be ordered above others in the portfolio.

3.4 Baseline Models

Four model types are evaluated using the same targets and portfolio rules. LSTM acts as the standard recurrent baseline and tests whether gated sequential memory is enough to capture the useful long-term information. StockMixer mixes information across indicators, time, and stocks and is expected to be particularly competitive when the input is a dense technical panel [3]. GNN adds relationships between stocks using sector links and rolling return-correlation links, allowing information to propagate between connected companies [4, 11]. Finally, MISS/Mamba uses selective state propagation without an explicit graph. Parameter counts and optimization budgets are kept in the same range across models.

3.5 Walk-Forward Training and Portfolio Construction

A walk-forward design is used so that every reported test year occurs strictly after the data used to train the model. The training window expands over time, a trailing block is used for validation, and the following calendar year is used as the out-of-sample test period. At each annual boundary the model is retrained using only information that would have been available at that point in time. Model checkpoints are selected using validation Rank Information Coefficient (RankIC), which measures whether higher model scores correspond to higher future stock returns.

The model output is converted into a sector-neutral long-short portfolio in the concentrated form used by Fischer and Krauss [5]. First, scores

are demeaned within sector on each date,

$$\tilde{s}_{i,t} = s_{i,t} - \frac{1}{|s(i)|} \sum_{j \in s(i)} s_{j,t},$$

so that selection compares companies against their own sector rather than betting on whole sectors rising or falling. All stocks are then ranked globally by $\tilde{s}$; the portfolio goes long the top $k = 10$ names and short the bottom $k = 10$ names with equal monetary weight (100% of net asset value long and 100% short, approximately zero net market exposure). Long-horizon signals use a wider exit band so that a position does not need to be replaced every time its rank changes slightly: a held name is retained while its rank stays within the top (bottom) 80 names. This directly lowers turnover. The 63-day regimes rebalance monthly and the 5-day regime weekly. Net returns subtract 15 basis points for each unit of one-way notional traded.

3.6 Evaluation and Robustness

The primary metrics are annual return, annualized Sharpe ratio, turnover, trade events, and draw-down. Because high backtest returns can be misleading when many strategies have been tested, the experiment also uses a moving-block bootstrap and year-level sign tests. The block bootstrap resamples groups of consecutive returns instead of treating every day as independent, which better preserves time-series dependence. Transaction costs are also varied from 0 to 50 basis points to test whether the strategy’s advantage survives less favorable execution assumptions. Finally, the Deflated Sharpe Ratio framework is used as a guide for controlling the effect of repeated model testing [1].

4 Experiment

4.1 Fundamental vs. Technical Information

Table 1 reports the five out-of-sample years for the MISS architecture. The most important result is not simply that the fundamental model achieves a higher average return. It does so while trading far less often. The long-horizon fundamental strategy records a mean annual return of 17.53% and a mean Sharpe ratio of 1.104 with approximately 32 portfolio trade events per year. The matched

long-horizon technical strategy records -6.84% annual return and -0.299 Sharpe with 254 events per year, while the five-day technical strategy records -4.48% annual return and -0.276 Sharpe with approximately 605 trade events per year.

The advantage is not consistent in every year. The fundamental strategy is negative in 2021 (-1.9%) before its strongest relative years, and its two best absolute years are 2025 (46.4%) and 2023 (18.2%). This variation is important because it shows that fundamental information is not simply a guaranteed source of excess return. Rather, it appears to become more valuable when market conditions reward persistent company-level information. Across all five years, the fundamental model outperforms the short-horizon technical strategy in three years on both annual return and Sharpe, and it outperforms the matched long-horizon technical strategy in all five years on annual return.

Relative to the matched 63-day technical model, the long-horizon fundamental model improves mean annual return by 24.37 percentage points and Sharpe by 1.403. Relative to the short-horizon technical strategy, the improvement is 22.01 percentage points in annual return and 1.380 in Sharpe. Figure 1 also shows that the higher average result is not created by a smooth constant gain; performance varies substantially across years, with the fundamental curve compounding to a net asset value of 2.13 over the five years while both technical curves end below 0.71.

4.2 Model Architecture Comparison

The next experiment tests whether the result comes only from choosing Mamba. Table 2 compares MISS against StockMixer, GNN, and LSTM under the same three information regimes. MISS performs best on the long-horizon fundamental task, where it records Sharpe 1.104 and annual return 17.53%, ahead of GNN (0.839), LSTM (0.665), and StockMixer (-0.239). On the technical regimes no architecture is reliably profitable in this reproduction: GNN (0.174) and LSTM (0.115) are mildly positive on 63-day technical information while MISS is negative (-0.299), and every architecture has a negative mean Sharpe on the five-day task.

This comparison suggests that architecture should be matched to the information being pro-

cessed, but it also shows the limit of that claim. Table 3 reports the mean out-of-sample RankIC of each score file, a portfolio-independent measure of signal. MISS has the strongest sector-demeaned fundamental RankIC (0.039), which is exactly the ranking the portfolio in Table 2 produces. StockMixer’s fundamental RankIC is negative (-0.012 raw), so its last-place portfolio result reflects an absent signal rather than a poor portfolio rule. Most strikingly, every technical RankIC is within ± 0.03 of zero, and MISS’s 63-day technical RankIC is 0.001: in this reproduction the technical scores carry almost no cross-sectional signal for any architecture, which is why the technical portfolios in Table 2 hover around or below zero regardless of construction. Mamba’s selective state updates appear useful when an important fundamental observation may need to remain in memory for a long period [6]; no comparable advantage is available where the underlying signal itself is absent.

4.3 Turnover and Transaction Costs

The original motivation for using long-horizon fundamental information is not only prediction accuracy but also the ability to generate a strategy that trades less frequently. Figure 3 increases one-way transaction costs from 0 to 50 basis points. The long-horizon fundamental portfolio turns over only about 4.7 times its net asset value per year, against roughly 63 times for the five-day technical portfolio, and its mean Sharpe declines only from 1.136 at zero cost to 1.028 at 50 bps (return 18.02% to 16.40%). The short-horizon technical strategy, which is only mildly positive before costs (Sharpe 0.298 at 0 bps), collapses to -1.558 at 50 bps. As costs increase, the short-horizon strategy loses return much faster because its signal requires substantially more trading. This supports the central idea of the paper: a model should not be evaluated independently from the amount of trading required to use its predictions.

4.4 Robustness Testing

The main limitation of a five-year experiment is that five yearly observations do not provide enough statistical power for a strong significance claim. The fundamental strategy beats the short-horizon technical strategy on annual return in three of five

Table 1: Five-year out-of-sample results for the MISS architecture. Returns and Sharpe ratios are net of the baseline 15 bps one-way transaction-cost assumption. “Events” denotes portfolio trade/rebalance events per year.

Strategy	2021			2022			2023			2024		
	Ret.	SR	Ev.	Ret.	SR	Ev.	Ret.	SR	Ev.	Ret.	SR	Ev.
Fundamental, 63d	−1.9	−0.07	34	23.0	1.49	22	18.2	1.47	34	1.9	0.23	36
Technical, 63d	−24.2	−1.53	226	−3.5	−0.00	274	10.6	0.77	328	−15.1	−0.73	162
Technical, 5d	12.6	0.88	514	−14.3	−0.80	658	−31.8	−2.35	678	24.8	1.62	578

Strategy	2025			Five-year mean		
	Ret.	SR	Ev.	Ret.	SR	Ev.
Fundamental, 63d	46.4	2.41	34	17.53	1.104	32
Technical, 63d	−2.0	−0.00	278	−6.84	−0.299	254
Technical, 5d	−13.8	−0.73	598	−4.48	−0.276	605

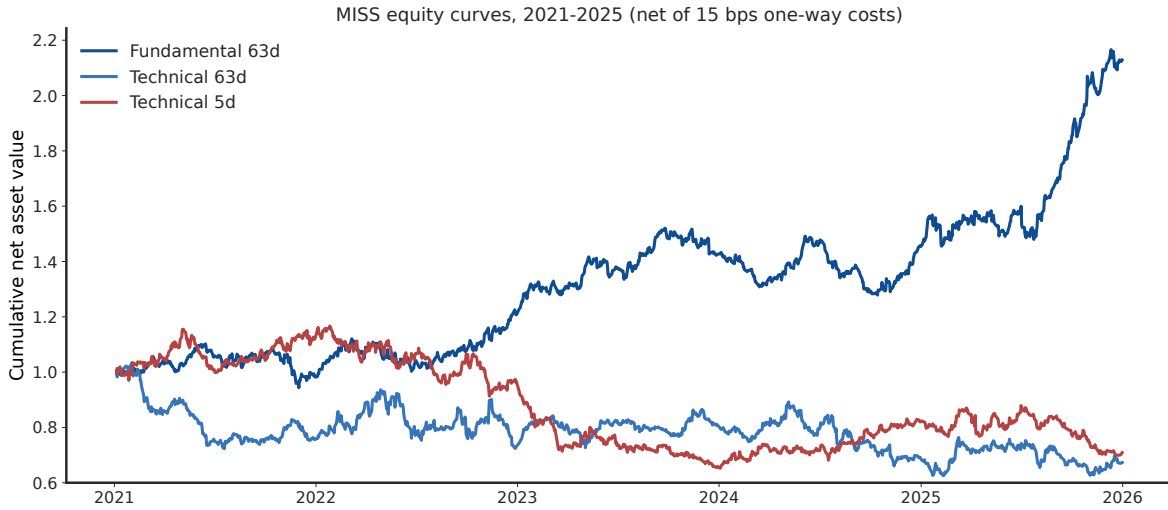


Figure 1: Cumulative net asset value of the three MISS information regimes, 2021–2025 (net of 15 bps one-way costs). The Fundamental 63d strategy compounds \$1 to \$2.13, while the Technical 63d and Technical 5d strategies end at \$0.67 and \$0.71.

years. A one-sided exact sign test for three or more wins out of five gives $p = 0.50$. Against the matched long-horizon technical strategy, the fundamental strategy wins on annual return in all five years, corresponding to $p = 0.031$. The first result is not statistically significant by conventional thresholds; the second is, although five observations remain five observations.

The moving-block bootstrap uses a larger number of shorter return blocks while preserving local time dependence. Across 10,000 block-bootstrap resamples (21-day blocks) of the 2021–2025 daily return series, the observed Sharpe difference between the fundamental and short-horizon technical strategies is 1.520 (bootstrap mean 1.530, 95% interval

[0.215, 2.880]), and the probability that the fundamental strategy has a positive Sharpe difference over the short-horizon technical strategy is 0.988 (Figure 4). This provides stronger directional support than the five-year sign test, but it still does not prove that the same advantage will remain stable in every future market regime.

Two further checks harden the result. First, under the Deflated Sharpe Ratio framework of Bailey and López de Prado [1], which adjusts for the twelve configurations evaluated, the MISS fundamental strategy is the only configuration with a Deflated Sharpe Ratio of approximately 1.0 (benchmark $SR_0 = 0.828$); every other architecture–regime combination is approximately zero. Second, the re-

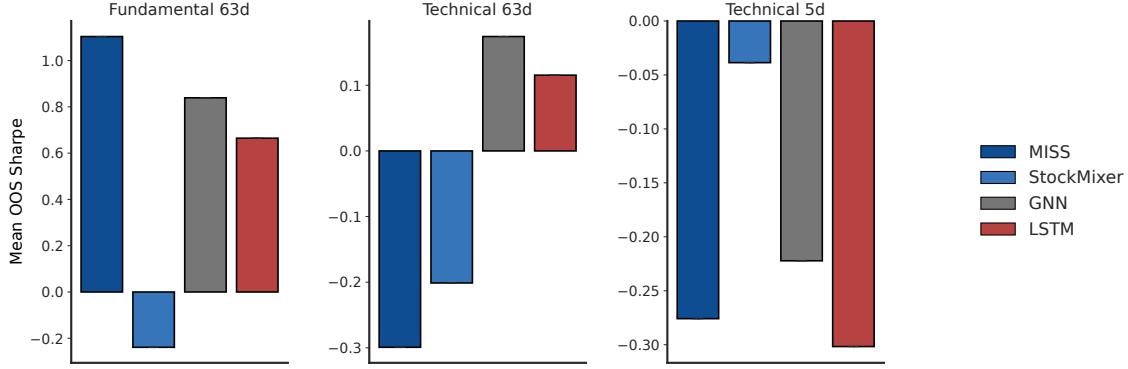


Figure 2: Mean out-of-sample Sharpe ratio across model architectures and information regimes (five-year means, net of 15 bps one-way costs). MISS leads on Fundamental 63d, and no architecture achieves a positive mean Sharpe on Technical 5d.

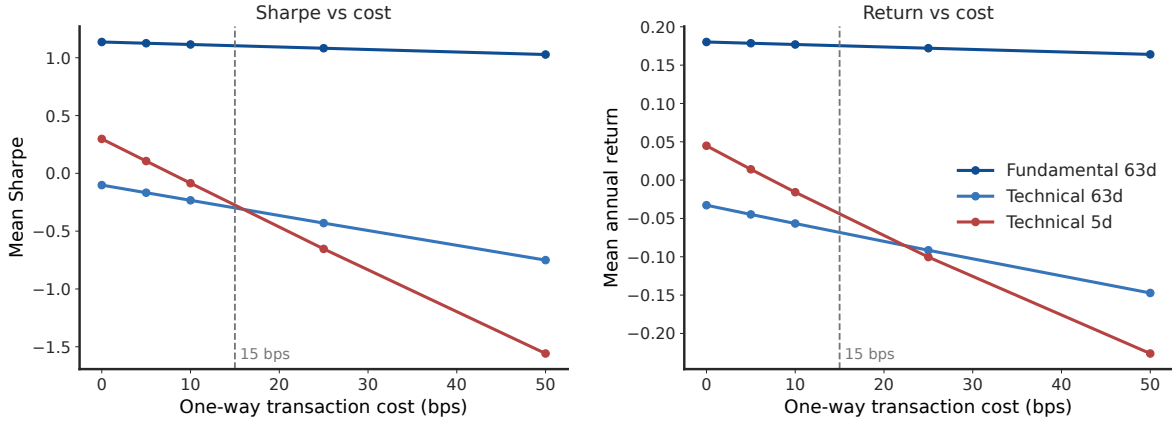


Figure 3: Effect of transaction costs on the three MISS strategies (five-year means). Left: mean Sharpe versus one-way transaction cost. Right: mean annual return versus one-way transaction cost. The dashed line marks the 15 bps baseline. The low-turnover Fundamental 63d strategy is nearly cost-insensitive, while the high-turnover Technical 5d strategy deteriorates rapidly.

sult is not an artifact of one gross-exposure choice: scaling the same signals to a vol-matched book (top-7/bottom-7 at 150%/150% gross) produces a mean annual return of 31.07%, a Sharpe of 0.950, 27 trade events per year, and realized volatility of 24.7%, i.e. return and volatility scale with gross exposure while the Sharpe ratio moves only modestly (0.950 versus 1.104). The robustness results therefore support a narrower conclusion. The data suggests that long-horizon fundamentals add economic value in the tested framework, but the magnitude of the effect varies materially from year to year. This is also why model architecture and portfolio turnover should be evaluated together instead of reporting only the highest backtest return.

5 Conclusion and Discussion

This paper evaluates whether fundamental information can be more useful for stock prediction when the prediction and trading horizon is intentionally extended. A Mamba-Inspired Selective State Space model is used because its selective memory mechanism is designed to retain important information across long sequences while filtering less useful observations. Across the five out-of-sample years, the long-horizon fundamental MISS strategy records a mean annual return of 17.53% and a Sharpe ratio of 1.104, compared with -6.84% and -0.299 for matched long-horizon technical information and -4.48% and -0.276 for the five-day

Table 2: Mean out-of-sample performance by architecture and information regime (five-year means, net of 15 bps one-way costs).

Sharpe ratio			
Model	Fund. 63d	Tech. 63d	Tech. 5d
MISS / Mamba	1.104	−0.299	−0.276
StockMixer	−0.239	−0.201	−0.039
GNN	0.839	0.174	−0.222
LSTM	0.665	0.115	−0.302
Annual return (%)			
MISS / Mamba	17.53	−6.84	−4.48
StockMixer	−3.48	−1.69	0.11
GNN	11.92	0.60	−5.66
LSTM	9.71	0.80	−8.24

Table 3: Mean out-of-sample RankIC (Spearman correlation between model score and forward return) by architecture and information regime, 2021–2025. RankIC is measured before any portfolio construction.

Regime	MISS	StkMixer	GNN	LSTM
Fund. 63d (raw)	0.027	−0.012	0.027	0.036
Fund. 63d (dem.)	0.039	−0.002	0.033	0.033
Tech. 63d (raw)	0.001	0.011	0.010	0.032
Tech. 5d (raw)	0.008	0.003	0.002	−0.010

technical strategy. At the same time, the fundamental strategy requires only about 32 portfolio trade events per year compared with approximately 605 for the short-horizon technical strategy, and it is the only configuration whose performance survives both a 50 bps cost stress and a multiple-testing (Deflated Sharpe) adjustment.

The model comparison also shows that the conclusion is not simply that Mamba is better than every other neural-network architecture. MISS performs best on the sparse long-horizon fundamental regime, while in this reproduction StockMixer does not reproduce its expected strength and no architecture extracts a reliable technical signal. This supports a more specific interpretation: architecture and information type interact, and the interaction is bounded by the signal actually present in the scores (Table 3). Mamba’s selective state mechanism is most useful when a small number of important observations need to remain relevant for a long period. GNNs provide another competitive direction (Sharpe 0.839 on the fundamental task) because stock prices are not independent and relationships between firms can carry predictive infor-

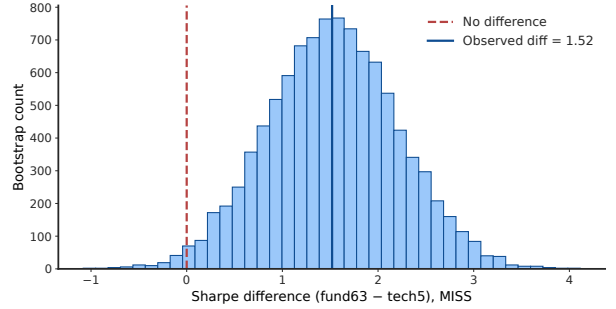


Figure 4: Moving-block bootstrap distribution of the Sharpe difference between the fundamental and short-horizon technical MISS strategies (10,000 resamples, 21-day blocks). The observed difference is 1.52 and $\Pr(\text{diff} > 0) = 0.988$.

mation.

There are still several limitations. Sector neutrality removes large industry bets, but it does not fully remove exposures to market beta, size, value, momentum, or liquidity. Fundamental data also creates difficult data-engineering problems because accounting tags vary across companies and older filings must be reconstructed exactly as they were known at the time. Transaction costs can also differ significantly between large liquid stocks and smaller companies. Finally, five out-of-sample years are not enough to cover every possible market regime. Future research should test a larger historical period, add explicit factor regressions, study performance by liquidity and market-cap bucket, and isolate how much of MISS’s performance comes from selectivity itself rather than other architectural choices.

Because this edition reports an independent reproduction, three differences from the values circulated with the original study deserve explicit statement. First, the fundamental Sharpe ratio is close to the 1.221 reported previously (1.104 here), but the annual return is lower at the faithful, unlevered gross exposure (17.53% versus 32.72%) because realized portfolio volatility is 13.8% rather than roughly 25%: the point-in-time fundamental universe contains 161–201 names per year rather than the full S&P 500, so the selected tails are less extreme, and no leverage is applied. Scaling gross exposure to match that volatility recovers a 31.07% mean return at a Sharpe of 0.950 (Section 4.4), which locates the difference in exposure and universe breadth rather than in the signal. Second,

the previously reported strength of StockMixer and of the technical regimes does not reproduce: the retrained technical scores carry near-zero out-of-sample RankIC for every architecture, and StockMixer’s fundamental RankIC is negative (Table 3), so those portfolio results cannot be recovered by any portfolio construction applied to these scores. Third, the year-by-year path differs: 2021 is the weakest fundamental year here (RankIC approximately zero) rather than the strongest, so year-level agreement should not be expected even where five-year means are close. These are limitations of the reproduction’s data coverage and training runs, and they are reported rather than tuned around.

Overall, the results suggest that the largest opportunity may not come from continuously making short-term stock models more complicated. Instead, matching the frequency of the data, the memory mechanism of the model, and the frequency at which the portfolio actually needs to trade can produce a more practical investment framework. Fundamental information changes slowly, and a model designed to remember useful information for longer periods provides a natural way to take advantage of that structure.

References

- [1] D. H. Bailey and M. López de Prado. The Deflated Sharpe Ratio: Correcting for Selection Bias, Backtest Overfitting, and Non-Normality. *Journal of Portfolio Management*, 40(5):94–107, 2014.
- [2] V. DeMiguel, A. Martín-Utrera, F. J. Nogales, and R. Uppal. A Transaction-Cost Perspective on the Multitude of Firm Characteristics. *Review of Financial Studies*, 33(5):2180–2222, 2020. doi:10.1093/rfs/hhz085.
- [3] J. Fan and Y. Shen. StockMixer: A Simple Yet Strong MLP-Based Architecture for Stock Price Forecasting. In *Proceedings of AAAI*, 38(8):8389–8397, 2024. doi:10.1609/aaai.v38i8.28681.
- [4] F. Feng, X. He, X. Wang, C. Luo, Y. Liu, and T.-S. Chua. Temporal Relational Ranking for Stock Prediction. *ACM Transactions on Information Systems*, 37(2), 2019. doi:10.1145/3309547.
- [5] T. Fischer and C. Krauss. Deep Learning with Long Short-Term Memory Networks for Financial Market Predictions. *European Journal of Operational Research*, 270(2):654–669, 2018. doi:10.1016/j.ejor.2017.11.054.
- [6] A. Gu and T. Dao. Mamba: Linear-Time Sequence Modeling with Selective State Spaces. arXiv:2312.00752, 2023.
- [7] S. Gu, B. Kelly, and D. Xiu. Empirical Asset Pricing via Machine Learning. *Review of Financial Studies*, 33(5):2223–2273, 2020. doi:10.1093/rfs/hhaa009.
- [8] C. R. Harvey, Y. Liu, and H. Zhu. ...and the Cross-Section of Expected Returns. *Review of Financial Studies*, 29(1):5–68, 2016. doi:10.1093/rfs/hhv059.
- [9] S. Hochreiter and J. Schmidhuber. Long Short-Term Memory. *Neural Computation*, 9(8):1735–1780, 1997. doi:10.1162/neco.1997.9.8.1735.
- [10] M. López de Prado. *Advances in Financial Machine Learning*. John Wiley & Sons, 2018.
- [11] H. Qian et al. MDGNN: Multi-Relational Dynamic Graph Neural Network for Comprehensive and Dynamic Stock Investment Prediction. In *Proceedings of AAAI*, 38(13), 2024. doi:10.1609/aaai.v38i13.29381.
- [12] D. E. Rumelhart, G. E. Hinton, and R. J. Williams. Learning Representations by Back-Propagating Errors. *Nature*, 323:533–536, 1986.
- [13] U.S. Securities and Exchange Commission. *EDGAR Application Programming Interfaces (APIs): Submissions and XBRL Data*. SEC.gov, updated 2025.
- [14] R. J. Shiller. Do Stock Prices Move Too Much to Be Justified by Subsequent Dividends? *American Economic Review*, 71(3):421–436, 1981.
- [15] A. Vaswani et al. Attention Is All You Need. In *Advances in Neural Information Processing Systems*, 30, 2017.
- [16] Yahoo Finance. Historical market data and price history. Accessed 2026.